॥ श्रीः ॥

लीलावती

श्रीभास्कराचार्यविरचिता

Līlāvatī of Bhāskarācārya

Bilingual Sanskrit–English Edition

Treatise on Arithmetic, c. 1150 CE

संस्कृत मूलपाठ [illegible]	⟷	**English Translation** with IAST Transliteration

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Preface / प्रस्तावना

लीलावती भास्कराचार्यस्य सिद्धान्तशिरोमणेः प्रथमः भागः। श्रीभास्करेण ११५० ई.पू.रचिता। लीलावती नाम तस्य कन्यायाः नाम अस्ति। इयं ग्रन्थः भारतीयगणितस्य सर्वाधिकप्रसिद्धपाठ्यपुस्तकम् अस्ति। पूर्णाङ्केषु, भिन्नेषु, अनुपातेषु, व्याजेषु, क्षेत्रफलेषु, घनफलेषु, श्रेणीषु च गणितविषयान् व्यवस्थापयति।

līlāvatī bhāskarācāryasya siddhāntaśiromaṇeḥ prathamaḥ bhāgaḥ | śrībhāskareṇa 1150 ī.pū.racitā | līlāvatī nāma tasya kanyāyāḥ nāma asti | iyāṃ granthaḥ bhāratīyagaṇitasya sarvādhikaprasiddhapāṭhyapustakam asti |

The *Līlāvatī* is the first section of Bhāskara II's monumental work *Siddhāntaśiromaṇi*, composed in 1150 CE. Named after Bhāskara's daughter, it is the most celebrated textbook of Indian arithmetic. The work covers the entire field of elementary mathematics: arithmetic operations with integers and fractions, rules of proportion, interest calculation, plane geometry, solid geometry, and combinatorial problems.

अस्य ग्रन्थस्य प्रस्तावने — अत्र गणितस्य समग्रं क्षेत्रम् उपस्थापितम् अस्ति। सङ्कलनम्, व्यवकलनम्, गुणनम्, भागहारः, वर्गः, वर्गमूलम्, घनः, घनमूलम् — एतेषां परिकर्मणां व्यवस्था, भिन्नपरिकर्म, त्रैराशिकम्, मिश्रप्रश्नाः, क्षेत्रव्यवहारः, खातव्यवहारः, चितिव्यवहारः, छायाव्यवहारः, कुट्टकव्यवहारः, अङ्कपाशः, श्रेणी चेत्यादयः अध्यायाः सन्ति।

asya granthasya prastāvane — atra gaṇitasya samagraṃ kṣetram upasthāpitam asti | saṅkalanaṃ, vyavakalanaṃ, guṇanaṃ, bhāgahāraḥ, vargaḥ, vargamūlam, ghanaḥ,ghanamūlam — eteṣāṃ parikarmaṇāṃ vyavasthā, bhinnaparikarma, trairāśikam, miśrapraśnāḥ, kṣetravyavahāraḥ, khātavyavahāraḥ, citivyavahāraḥ, chāyāvyavahāraḥ, kuṭṭakavyavahāraḥ, aṅkapāśaḥ, śreṇī cetyādayaḥ adhyāyāḥ santi |

The present edition reproduces the Sanskrit text in the left column with an English translation and IAST transliteration in the right column. Key technical terms are preserved in their original Sanskrit form with explanatory footnotes.

मङ्गलाचरण — Invocatory Verses

लीलागुल्लक्षोल्कालव्यालविलसिने ।
गणेशाय नमो नीलकमलामलकान्तये ॥
॥ १ ॥

Līlāgullakṣolkāvyālavilasine
Gaṇeśāya namo nīlakamalāmalakāntaye

Salutation to Gaṇeśa, the playfully sportive one, having the brightness of the rising sun, and possessing the radiance of a blue lotus.

अवतार पुरा पुराणेषां यदुकुले जगतीहितेहिने ।
जयति सोऽयमिमां सकलां लीलावतीं मामपि सिद्धमहीपतिः ॥

avatāra purā purāṇeṣām̐ yadukule jagatīhitehine |
jayati so'yamimām̐ sakalām̐ līlāvatīm̐ māmapi siddhamahīpatiḥ ||

In ancient times, he who took incarnation in the Yadu clan for the welfare of the world — may that same king, the lord of the earth, be victorious, who has made me compose this entire *Līlāvatī*.

भास्करचितयमथा सिद्धान्तशिरोमणिप्रख्याः ।
तदयं कृताऽनन्ये व्याकृतया मन्मठे नियतं ॥
॥ १५ ॥

Bhāskaracitayamathā siddhāntaśiromaṇiprakhyāḥ
Tadayaṃ kṛtā'nye vyākṛtayā manmaṭhe niyataṃ

This *Līlāvatī*, composed by Bhāskara, is the foremost part of the *Siddhāntaśiromaṇi*. This has been composed by none other than him, and is now set down by me.

1 परिभाषामकरण — Definitions and Units of Measurement

पादोनग्रानकतुल्यष्टकैः समतुल्यैः कथितोऽत्र सेरः ॥
मणाभिधानं व्युग्मैः सैरेधार्यादितोल्येषु तुष्टकसङ्ख्या ॥ १ ॥

pādonagrānakatulyaṣṭakaiḥ samatulyaiḥ kathito'tra seraḥ ||
maṇābhidhānaṃ vyugmaiḥ sairedhāryāditolyeṣu tuṣṭakasaṅkhyā || 1 ||

A *ser* is equal to eight *pāda* or equal to eight grains. A *maṇa* is twice the amount of a *ser* in weighing gold, etc. *[The ser and maṇa were traditional Indian units of weight. 1 maṇa = 2 ser.]*

द्वयकद्विसहस्रैर्घटकैः सैरस्तैः पञ्चभिः स्याद्घटिका च ताभिः ॥
मणोऽष्टमिर्त्वालमगीरशाहकृतास्त्र सङ्ख्या निजराज्यपृष्ठे ॥ २ ॥

dvayakadvisahasrairghaṭakaiḥ sairastaiḥ pañcabhiḥ syāddhaṭikā ca tābhiḥ ||
maṇo'ṣṭamirtvālamagīraśāhakṛāstra saṅkhyā nijarājyapṛṣṭhe || 2 ||

Two thousand *ghaṭakas* make a *ser*; five of these *sers* make a *ghaṭikā*. The weight of one *maṇa* in this kingdom (of the author's patron) follows this system.

1.1 शोभाः कालादिपरिभाषा — Definitions of Time

शोभाः कालादिपरिभाषा लोकतः प्रसिद्धा देया ॥

śobhāḥ kālādiparibhāṣā lokataḥ prasiddhā deyā ||

The definitions of time and other measures are well-known in the world. The traditional Indian system of time measurement is given as:

- 60 *vipalas* = 1 *pala*
- 60 *palas* = 1 *ghaṭikā* (24 minutes)
- 60 *ghaṭikās* = 1 day and night
- 15 days and nights = 1 *pakṣa* (fortnight)
- 2 *pakṣas* = 1 month
- 12 months = 1 year

इति परिभाषा — Thus end the definitions.

2 सङ्ख्याप्रकरण — Arithmetic Operations on Whole Numbers

This chapter covers the fundamental operations of arithmetic: addition (*saṅkalita*), subtraction (*vyavakalita*), multiplication (*guṇana*), division (*bhājana*), squaring (*varga*), square roots (*vargamūla*), cubing (*ghana*), and cube roots (*ghanamūla*).

2.1 सङ्कलित व्याकलित — Addition and Subtraction

सूत्रम्: यथोक्तं परिकल्प्य गणितं कुर्यात्।

Rule: As taught, having set up the numbers [aligned by place value], one should compute.

2.2 गुणकारकरणसूत्र — Multiplication

सूत्रम्: समद्विघातः कृतिः उच्यते—

Rule: A product of two equal numbers is called a *kṛti* (square). The multiplication of two different numbers is called *guṇana*.

अथ रूपयोऽन्त्यवर्गो द्विगुणान्त्यनिघ्नः...
एकस्यांकस्य वर्गः अन्तिमस्य अङ्कस्य गुणनम्। द्विगुणान्त्यनिघ्नः = द्विगुणः अन्तिमः अङ्कः अन्येन गुणितः।

atha rūpayo'ntyavargo dviguṇāntyanighnaḥ...
ekasyāṃkasya vargaḥ antim asya aṅkasya guṇanam |
Method of Multiplication: The square of the last figure, and twice the last multiplied by the preceding, and so on — this is the method for squaring.
Example: To multiply 135 by 12:

$$135 \times 12 = 135 \times 10 + 135 \times 2 = 1350 + 270 = 1620$$

2.3 भागहारकरणसूत्र — Division

सूत्रम्: भक्तः क्षेपणात् पुनः पुनरवाप्तिर्भागहारः।

Rule: Division is the repeated subtraction of the divisor from the dividend until the remainder is less than the divisor.

Division is performed by repeated subtraction. The commentary gives detailed worked examples. The quotient (*labdhi*) is the number of subtractions, and what remains is the remainder (*śeṣa*).

2.4 वर्गीकरणसूत्र — Squaring

सूत्रम्: समद्विघातः कृतिः उच्यते—

Rule: The product of two equal numbers is called a *kṛti* (square).

खण्डाभ्यां वा हतो राशिरिष्टः खण्डचतुष्कयुक् ॥
अस्य अर्थः — राशिं द्वयोः खण्डयोः योगेन गुणित्वा तयोरन्तरवर्गेण सह योजयेत्।

khaṇḍābhyāṃ vā hato rāśiriṣṭaḥ khaṇḍacatuṣkayuk ||
asya arthaḥ — rāśiṃ dvayoḥ khaṇḍayoḥ yogenaguṇitvā tayorantaravargeṇa saha yojayet |
Method 1: Direct multiplication $n \times n$
Method 2: Using the formula $(a + b)^2 = a^2 + 2ab + b^2$
Method 3: Using the formula $n^2 = (n + a)(n - a) + a^2$
Or, the number multiplied by the component parts of itself, combined with the square of the difference...

2.5 वर्गमूलकरणसूत्र — Square Root Extraction

सूत्रम्: स्थाप्योऽन्त्यवर्गो द्विगुणोऽन्त्यनिघ्नः स्यादपरिच्छेदः।

Rule: Set down the number. Find the largest square in the leftmost group. This gives the first digit of the root. Subtract its square, bring down the next group, divide by twice the current root, and repeat.

अथ स्थाप्योऽन्त्यवगो द्विगुणान्त्यनिघ्नः॥
स्यादपरिच्छेदः तथा परेऽभ्यस्तयुक्त्वान्त्यमपि राशिम् ॥

atha sthāpyo'ntyavago dviguṇāntyanighnaḥ ||
syādaparicchedaḥ tathā pare'bhyastayuktvāntyamapi rāśim ||

Example: Find $\sqrt{65536}$
Step 1: Group the digits from right: $\overline{6}\,\overline{55}\,\overline{36}$
Step 2: Largest square ≤ 6 is 4 = 2^2. First digit: 2.
Step 3: Remainder 2, bring down 55 $\rightarrow$ 255.
Step 4: $255 \div (2 \times 20) = 255 \div 40 \approx 5$.
Step 5: Verify: $45 \times 5 = 225$. Remainder 30.
Step 6: Bring down 36 $\rightarrow$ 3036.
Step 7: $3036 \div (2 \times 250) = 3036 \div 500 \approx 6$.
Step 8: Verify: $506 \times 6 = 3036$. Remainder 0.
Therefore $\sqrt{65536} = 256$.

2.6 घनकरणसूत्र — Cubing

सूत्रम्: समत्रिघातः घनः उच्यते—

Rule: The product of a number multiplied three times by itself is called a *ghana* (cube).

Methods for cubing:
Method 1: Direct multiplication $n \times n \times n$
Method 2: Using $(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$

2.7 घनमूलकरणसूत्र — Cube Root Extraction

The method follows a similar place-value decomposition as square root extraction, but using cubes of digits.
Example: $\sqrt[3]{1728} = 12$, since $12^3 = 1728$.

3 भिन्नपरिकर्म — Operations with Fractions

3.1 जातिचतुष्टय — Four Classes of Fractions

Bhāskara defines four types of fractions (*jāticatuṣṭaya*):

1. भिन्न (*bhinna*) — Simple fraction, e.g., $\frac{1}{2}$
2. प्रभाग (*prabhāga*) — Part of a part, e.g., $\frac{1}{2}$ of $\frac{1}{3}$
3. भागानुबन्ध (*bhāgānubandha*) — Fraction combined with an integer, e.g., $2 + \frac{1}{3}$
4. भागापवाह (*bhāgāpavāha*) — Fraction less than an integer, e.g., $2 - \frac{1}{3}$

3.2 भिन्नगुणन भिन्नभाग — Multiplication and Division of Fractions

सूत्रम्: नरघ्ननरके क्रमशो गुणयेत् तद् भागहारकैः ॥

Rule: Multiply the numerator by the numerator and the denominator by the denominator for multiplication; for division, invert and multiply.

Multiplication: $\frac{a}{b} \times \frac{c}{d} = \frac{a \times c}{b \times d}$

Division: $\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c} = \frac{a \times d}{b \times c}$

Example: $\frac{2}{3} \times \frac{5}{7} = \frac{10}{21}$

Example: $\frac{2}{3} \div \frac{5}{7} = \frac{2 \times 7}{3 \times 5} = \frac{14}{15}$

4 त्रैराशिक — The Rule of Three

The *trairāśika* (Rule of Three) is one of the most important topics in Indian arithmetic. It solves problems of the form:

$$\text{“If } A \text{ gives } B\text{, what does } C \text{ give?”} \quad \text{Answer: } \frac{B \times C}{A}$$

4.1 इच्छाकर्मप्रकार — Direct Rule of Three

सूत्रम्: प्रमाणफलमिच्छाफलं चेत् कुर्यात्।

Rule: Multiply the *phala* (fruit/result) by the *icchā* (desire) and divide by the *pramāṇa* (measure).

प्रमाणं फलमिच्छा च त्रिविधा राशयः स्मृताः ॥
प्रमाणेन यद् लभ्यते तत् फलम्। इच्छया यद् ज्ञातुम् इष्टम् तत् इच्छाफलम्।

pramāṇaṃ phalamicchā ca trividhā rāśayaḥ smṛtāḥ ||

Three quantities are remembered: the *pramāṇa* (measure), the *phala* (fruit), and the *icchā* (desire).
Example: If 5 books cost 60 rupees, what do 8 books cost?

$$\text{Price} = \frac{60 \times 8}{5} = 96 \text{ rupees}$$

4.2 व्यस्तत्रैराशिक — Inverse Rule of Three

सूत्रम्: व्यस्ते विपरीतं कुर्यात्।

Rule: In the inverse case, do the reverse [multiply measure by fruit and divide by desire].

When the relationship is inverse (more workers take less time, etc.):

$$\text{Result} = \frac{B \times A}{C}$$

Example: If 8 workers complete a job in 15 days, how many days will 12 workers take?

$$\text{Days} = \frac{8 \times 15}{12} = 10 \text{ days}$$

4.3 पञ्चराशिक सप्तराशिक — Extended Rules

Extended rules for compound proportions:

- पञ्चराशिक (*pañcarāśika*, Five-fold rule): Two intermediate terms
- सप्तराशिक (*saptarāśika*, Seven-fold rule): Three intermediate terms
- नवराशिक (*navarāśika*, Nine-fold rule): Four intermediate terms

5 मिश्रप्रकरण — Mixed Problems

5.1 शैलाराशिकविधि — Rule of Alligation

मिश्रणे विविधद्रव्याणां मूल्यानां समासेन...
सुवर्णादीनां मिश्रणे कलर्णज्ञानाय एषः प्रकारः।

miśraṇe vividhadravyāṇāṃ mulyānāṃ samāsena...
suvarṇādīnāṃ miśraṇe kalaṇajñānāya eṣaḥ prakāraḥ |

Problems involving the mixing of different quantities at different prices or qualities.
Example: If rice of quality A costs 8 rupees/kg and rice of quality B costs 12 rupees/kg, in what ratio should they be mixed to get a mixture worth 10 rupees/kg?

$$\text{Ratio} = (12 - 10) : (10 - 8) = 2 : 2 = 1 : 1$$

5.2 वापीपूरणप्रकार — Filling of Tanks

वापी पूरयितुं नलकैः कालज्ञानाय सूत्रम् ॥

vāpī pūrayituṃ nalakaiḥ kālajñānāya sūtram ||

Example: A tank has three pipes. The first can fill it in 5 hours, the second in 6 hours, and the third can empty it in 4 hours. If all three are opened together, how long will it take to fill the tank?

$$\text{Net fill rate} = \frac{1}{5} + \frac{1}{6} - \frac{1}{4} = \frac{12 + 10 - 15}{60} = \frac{7}{60}$$

Time to fill = $\frac{60}{7} = 8\frac{4}{7}$ hours.

5.3 कथाविकयविधि — Exchange of Goods

वस्तूनां विनिमये मूल्यसमत्वेन परस्परगुणनम् ॥

vastūnāṃ vinimaye mūlyasamatvena paras-paraguṇanam ||

Example: If 3 mangoes can be exchanged for 4 oranges, and 5 oranges for 2 apples, how many apples can be obtained for 15 mangoes?

$$15 \text{ mangoes} = 15\times\frac{4}{3} = 20 \text{ oranges} = 20\times\frac{2}{5} = 8 \text{ apples}$$

6 क्षेत्रव्यवहार — Plane Geometry

क्षेत्रे व्यवहारः —
॥ ॥

kṣetre vyavahāraḥ

This chapter deals with the calculation of areas of various plane figures and the properties of right triangles.

6.1 क्षेत्रसाधारणसूत्र — General Rule for Areas

सूत्रम्: दैर्घ्यमायतं विस्तारं च तद् दैर्घ्यविस्तारयोगजं क्षेत्रम्।

Rule: The area of a rectangle is length multiplied by breadth. The area of a figure is the product of its length and width.

6.2 भुजकोटिकर्णनयन — The Pythagorean Theorem

भुजे द्वादशके यो यो कोटिकर्णविनेकथा ॥
प्रकाराभ्यां वद विप्र तो तावकर्णीगतौ ॥ ५ ॥
॥ ५ ॥

bhuje dvādaśake yo yo koṭikarvavinekathā ||
prakārābhyāṃ vada vipra to tāvakarṇīgatāu || 5 ||

O learned Brahmin! The sides (*bhujā*) being 12, tell me quickly the hypotenuse (*karṇa*) and perpendicular (*koṭi*) by both methods.

इष्टयोराहतिर्द्वयोः कोटिवर्गान्तरं भुजः ॥
कृतियोगस्तयोरेवं कर्णस्त्वकरणीगतः ॥ ८ ॥
॥ ८ ॥

iṣṭayorāhatirdvayoḥ koṭivargāntaraṃ bhujaḥ ||
kṛtiyogastayorevaṃ karṇastvakaraṇīgataḥ || 8 ||

The product of the two desired [sides] is the difference of the squares of the *koṭi* and *bhujā*. The sum of the squares of those two gives the *karṇa*.

The Three Sides of a Right Triangle:

- **Base** (भुजा *bhujā*): The horizontal side
- **Perpendicular** (कोटि *koṭi*): The vertical side
- **Hypotenuse** (कर्ण *karṇa*): The diagonal side

The rule states:

$$\text{karṇa}^2 = \text{bhujā}^2 + \text{koṭi}^2$$

This is the famous Pythagorean theorem, known in India as the *Śulba-sūtra* theorem.
Example: Find the hypotenuse when base = 3 and perpendicular = 4:

$$\text{karṇa} = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$$

Example from the text: For base = 12, perpendicular = 5:

$$\text{karṇa} = \sqrt{12^2 + 5^2} = \sqrt{144 + 25} = \sqrt{169} = 13$$

Example: For base = 9, perpendicular = 12:

$$\text{karṇa} = \sqrt{9^2 + 12^2} = \sqrt{81 + 144} = \sqrt{225} = 15$$

6.3 कोटिकर्णज्ञान — Finding the Koṭi

When the hypotenuse and one side are known:

$$\text{koṭi} = \sqrt{\text{karṇa}^2 - \text{bhujā}^2}$$

Example: karṇa = 85, bhujā = 36:

$$\text{koṭi} = \sqrt{85^2 - 36^2} = \sqrt{7225 - 1296} = \sqrt{5929} = 77$$

Verification: $36^2 + 77^2 = 1296 + 5929 = 7225 = 85^2$ ✓

6.4 वृत्तक्षेत्र — Circle

सूत्रम्: व्यासवर्गपादं भुजवर्गस्य पञ्चगुणस्य चतुर्थांशः वृत्तफलम्।

Rule: The area of a circle is one-fourth the square of the diameter, or equivalently: the square of the radius multiplied by the ratio of circumference to diameter.

$$A = \frac{d^2}{4} \times \pi \approx \frac{d^2}{4} \times \frac{3927}{1250} \text{ (Bhāskara's refined value)}$$

Bhāskara uses $\pi \approx \frac{22}{7}$ for rough calculations and $\pi \approx \frac{3927}{1250} = 3.1416$ for more precise work.

7 खातव्यवहार — Excavations

खातव्यवहारः —
॥ ॥

khātavyavahāraḥ

This chapter deals with calculations related to excavations, trenches, and tanks.

खातस्य फलं दीर्घं विस्तारं गभीरता च तेषां गुणनफलम् ॥

khātasya phalaṃ dīrghaṃ vistāraṃ gabhīratā ca teṣāṃ guṇanaphalam ||

Problem: An excavation is to be made with length 30, breadth 20, and depth 8. Find the volume of earth to be removed.

$$V = l \times b \times d = 30 \times 20 \times 8 = 4800 \text{ cubic units}$$

7.1 सच्छिद्रखात — Excavations with Sloping Sides

For excavations with sloping sides, the *śāṇi* (average) method is used:

$$V = \frac{A_1 + A_2 + A_3 + A_4}{4} \times \text{length}$$

where A_1, A_2, A_3, A_4 are the cross-sectional areas at different depths.

8 चिति व्यवहार — Piles of Bricks

चितिव्यवहारे स्तूपचितीनां समचितीनां फलं यथासङ्ख्यम् ॥

citivyavahāre stūpacitīnāṃ samacitīnāṃ phalaṃ yathāsaṅkhyam ||

This chapter deals with stacking problems and the calculation of volumes of piles.
Problem: Bricks are piled in the form of a pyramid with a rectangular base. The base has 10 bricks along one side and 6 along the other. The pile is 5 layers high. Find the total number of bricks.
Solution: Using the formula for a pyramidal pile:

$$N = \frac{n(n+1)(2n+1)}{6} \text{ (for square pyramids)}$$

For rectangular piles, the sum is computed layer by layer.

9 छायाव्यवहार — Shadow Problems

छायाव्यवहारकथन —
॥ ॥

chāyāvyavahārakathana

This chapter uses the properties of similar triangles to solve problems involving shadows, heights of objects, and distances.

सूत्रम्: छायाग्रदूरं द्विजोत्कर्षं कुर्यात्।

Rule: The distance to the tip of the shadow multiplied by the height of the object, divided by the length of the shadow, gives the height of the light source.

दीपस्य स्तम्भस्य ऊर्ध्वे दीपः स्थापितः। तस्य छायायां पुरुषः स्थितः।

dīpasya stambhasya ūrdhve dīpaḥ sthāpitaḥ | tasya chāyāyāṃ puruṣaḥ sthitaḥ |

Problem: A lamp is placed on a pillar. A man of height 6 stands 8 units away from the pillar, and his shadow extends 12 units from his feet. Find the height of the lamp.
Solution: Using similar triangles:

$$\frac{h}{6} = \frac{8+12}{12} = \frac{20}{12}$$

$$h = 6 \times \frac{20}{12} = 10 \text{ units}$$

10 कुट्टकव्यवहार — The Pulverizer

कुट्टकव्यवहारः —
॥ ॥

kuṭṭakavyavahāraḥ

The *kuṭṭaka* (pulverizer) is the method for solving linear Diophantine equations of the form $ax + by = c$, where a, b, and c are given integers, and x, y are unknown integers to be found.

सूत्रम्: द्विच्छेदः कुट्टकः—

Rule: The pulverizer is the division of two (numbers), producing a quotient and remainder, used to find the GCD through the Euclidean algorithm.

कुट्टके स्थिरकुट्टकं प्रवर्तते। द्वयोः राश्योः महत्तमं समवायपदं ज्ञात्वा...
कुट्टकविधिः युक्लिडीयकलनविधिना सम्बद्धः अस्ति।

kuṭṭake sthirakuṭṭakaṃ pravartate | dvayoḥ rāśyoḥ mahattamaṃ samavāyapadaṃ jñātvā... ||

Problem: Find x and y such that $121x + 17 = y$
Solution: Using the kuṭṭaka method:
Step 1: Apply the pulverizer to 121 and 17:

$$\begin{aligned} 121 &= 7 \times 17 + 2 \\ 17 &= 8 \times 2 + 1 \end{aligned}$$

Step 2: Work backwards:

$$\begin{aligned} 1 &= 17 - 8 \times 2 \\ &= 17 - 8 \times (121 - 7 \times 17) \\ &= 57 \times 17 - 8 \times 121 \end{aligned}$$

So $x = 8$, $y = 121 \times 8 + 17 = 985$ gives a solution.

11 अङ्कपाश — Combinatorial Problems

अङ्कपाशः —
॥ ॥

aṅkapāśaḥ

This chapter contains problems related to permutations and combinations, and number puzzles. The Sanskrit term *aṅkapāśa* literally means "net of digits" — the entanglement of numbers.

11.1 अङ्कपाशविधि — Digit Arrangements

एकादशअङ्कैः कति विभिन्नसंख्याः रचयितुं शक्याः ?

ekādaśaṅkaiḥ kati vibhinnasaṅkhyāḥ racayituṃ śakyāḥ ?

Problem: Using the digits 1, 2, 3, 4, how many different 4-digit numbers can be formed?
Solution: $4! = 24$ different numbers.
This is the fundamental counting principle — each position can be filled by any of the remaining digits.

11.2 अङ्कपाशमें विशेषाविधि — Special Methods

एका संख्या त्रिगुणीकृता सप्तभक्ता द्विशेषं लभते।

ekā saṅkhyā triguṇīkṛtā saptabhaktā dviśeṣaṃ labhate |

Problem: A certain number, when multiplied by 3 and divided by 7, gives a remainder of 2. Find the number.
This is equivalent to solving:

$$3x \equiv 2 \pmod{7}$$

Since $3 \times 5 = 15 \equiv 1 \pmod{7}$, the multiplicative inverse of 3 mod 7 is 5.

$$x \equiv 2 \times 5 = 10 \equiv 3 \pmod{7}$$

So $x = 3 + 7k$ for any integer k. The smallest positive solution is $x = 3$.

12 श्रेणी — Progressions

श्रेणी —

॥ ॥

śreṇī

The *śreṇī* (progression) chapter covers arithmetic and geometric progressions.

12.1 समश्रेणी — Arithmetic Progression

समश्रेण्यां प्रथमपदं चयं गच्छं च ज्ञात्वा सर्वं ज्ञायते ॥

समश्रेण्यां (□□□□□□□□□□ □□□□□□□□□□□) प्रथमं पदम् a, चयः (□□□□□□ □□□□□□□□□□) d, गच्छः (□□□□□□ □□ □□□□□□) n चेत् —

samaśreṇyāṃ prathamapadaṃ cayaṃ gacchaṃ ca jñātvā sarvaṃ jñāyate ||

For an arithmetic progression with first term a, common difference d, and n terms:

- n-th term: $a_n = a + (n-1)d$
- Sum of n terms: $S_n = \frac{n}{2}[2a+(n-1)d] = \frac{n}{2}(a+l)$

where l is the last term.

12.2 गुणोत्तरश्रेणी — Geometric Progression

गुणोत्तरश्रेण्यां प्रथमपदं गुणोत्तरं च ज्ञात्वा सर्वं ज्ञायते ॥

guṇottaraśreṇyāṃ prathamapadaṃ guṇottaraṃ ca jñātvā sarvaṃ jñāyate ||

For a geometric progression with first term a and common ratio r:

- n-th term: $a_n = a \cdot r^{n-1}$
- Sum of n terms: $S_n = a \cdot \frac{r^n-1}{r-1}$

The Chessboard Problem:

Problem: A king offers a reward of 1 grain of rice on the first square of a chessboard, 2 on the second, 4 on the third, and so on, doubling each time. Find the total grains on all 64 squares.

Solution:

$$S_{64} = 1 + 2 + 4 + \cdots + 2^{63} = \frac{2^{64}-1}{2-1} = 2^{64} - 1$$
$$= 18{,}446{,}744{,}073{,}709{,}551{,}615 \text{ grains}$$

Appendix: Famous Problems from the Līlāvatī

The Lotus Problem (कमलसमस्या)

कमलस्य पत्रं जलात् ऊर्ध्वं स्थितम्। वायुना पीडितं जलस्य अन्तः गतम्।
॥ ॥

kamalasya patraṃ jalāt ūrdhvaṃ sthitam | vāyunā pīḍitaṃ jalasya antaḥ gatam |

A lotus flower stands x units above the water surface. When blown by the wind, it sinks d units from its original vertical position. Find the depth of the water.

Solution: If the lotus is x above the water, the stem length equals the depth h plus x. When blown, the stem forms the hypotenuse of a right triangle:

$$(h + x)^2 = h^2 + d^2$$

$$h^2 + 2hx + x^2 = h^2 + d^2$$

$$h = \frac{d^2 - x^2}{2x}$$

[This is a classic application of the Pythagorean theorem (bhujā-koṭi-karṇa relation).]

The Bee Problem (भ्रमरसमस्या)

भ्रमराणां पञ्चमांशः कुटजपुष्पे तृतीयांशः शिलीन्ध्रपुष्पे च ॥
अन्तरस्य त्रिगुणं क्रातपुष्पे एकः चन्दने विहरति ॥
॥ ॥

bhramarāṇāṃ pañcamāṃśaḥ kuṭajapuṣpe tṛtīyāṃśaḥ śilīndhrapuṣpe ca ||
antarasya triguṇaṃ krātapuṣpe ekaḥ candane viharati ||

A fifth of a swarm of bees settled on a *kuṭaja* flower, a third on a *śilīndhra* flower. Three times the difference between these two groups flew over a *krāta* flower. One bee was left flying about, attracted by the fragrance of a jasmine and a pandanus. How many bees were in the swarm?

Solution: Let x be the total number of bees.

$$\frac{x}{5} + \frac{x}{3} + 3\left(\frac{x}{3} - \frac{x}{5}\right) + 1 = x$$

$$\frac{x}{5} + \frac{x}{3} + 3 \cdot \frac{2x}{15} + 1 = x$$

$$\frac{3x + 5x + 6x}{15} + 1 = x \implies \frac{14x}{15} + 1 = x \implies x = 15$$

Answer: 15 bees. *[This is one of the most famous problems from the Līlāvatī, demonstrating skill in working with fractions and linear equations.]*

The Peacock and the Snake (मयूरसर्पकथा)

मयूरः स्तम्भे स्थितः सर्पः तस्य पादात् दूरे स्थितः ॥
मयूरः सर्पं ग्रहीतुं पतति समदूरी क्षिप्त्वा ॥
॥ ॥

mayūraḥ stambhe sthitaḥ sarpaḥ tasya pādāt dūre sthitaḥ ||
mayūraḥ sarpaṃ grahītuṃ patati samadūrī kṣiptvā ||

A peacock is perched on a pillar of height P. A snake is at distance D from the base of the pillar. The peacock pounces on the snake in a straight line, covering equal distances down the pillar and along the ground. Find where they meet.

Solution: Let the meeting point be at distance x from the base of the pillar. The peacock flies in a straight line, and the problem states it covers equal vertical and horizontal distances:

$$P - x = \sqrt{x^2 + D^2}$$

Squaring both sides:

$$(P - x)^2 = x^2 + D^2 \implies P^2 - 2Px + x^2 = x^2 + D^2$$

$$x = \frac{P^2 - D^2}{2P}$$

[This combines the Pythagorean theorem with the concept of equal distances. The problem is notable for its elegant reduction to a simple algebraic expression.]

The Swallow Problem (हंससमस्या)

Problem: Swallows are sitting on the branches of a tree. If each branch has 3 swallows, one swallow is left over. If each branch has 4 swallows, one branch is left empty. How many swallows and branches are there?

Solution: Let s = number of swallows, b = number of branches.

$$s = 3b + 1$$
$$s = 4(b - 1) = 4b - 4$$

Equating: $3b + 1 = 4b - 4$, so $b = 5$ and $s = 16$.
Answer: 16 swallows and 5 branches. *[This is a classic example of a system of linear equations with a unique solution. It is equivalent to finding x such that $x \equiv 1 \pmod 3$ and $x \equiv 0 \pmod 4$.]*

The Pearl Necklace Problem (मुक्ताहारसमस्या)

मुक्ताहारे मुक्ताः स्रंसन्ति। एकैकस्य मुक्तायाः मूल्यं ज्ञात्वा...
॥ ॥

muktāhāre muktāḥ sraṃsanti | ekaikasya muktāyāḥ mūlyaṃ jñātvā...

A string of pearls breaks. One third of the pearls fall to the ground, one fifth remain on the bed, one sixth are saved by the young woman, one tenth are caught by her lover, and six pearls remain on the string. How many pearls were there originally?

Solution: Let x be the total number of pearls.

$$\frac{x}{3} + \frac{x}{5} + \frac{x}{6} + \frac{x}{10} + 6 = x$$

The LCD of 3, 5, 6, 10 is 30:

$$\frac{10x + 6x + 5x + 3x}{30} + 6 = x \implies \frac{24x}{30} + 6 = x$$

$$6 = x - \frac{4x}{5} = \frac{x}{5} \implies x = 30$$

Answer: 30 pearls. *[This problem demonstrates the use of finding a common denominator and solving linear equations with fractions — a common practical problem type in the Līlāvatī.]*

Glossary of Technical Terms

संस्कृत	IAST	Meaning
व्यवहार	vyavahāra	Procedure, operation, practice
गुणन	guṇana	Multiplication
भागहार	bhāgahāra	Division
क्षेत्र	kṣetra	Field, area, plane figure
राशि	rāśi	Quantity, number, heap
सङ्कलन	saṅkalana	Addition
व्यवकलन	vyavakalana	Subtraction
वर्ग	varga	Square
वर्गमूल	vargamūla	Square root
घन	ghana	Cube
घनमूल	ghanamūla	Cube root
भिन्न	bhinna	Fraction
प्रमाण	pramāṇa	Measure (in Rule of Three)
फल	phala	Fruit, result
इच्छा	icchā	Desire, requisition
त्रैराशिक	trairāśika	Rule of Three
करण	karaṇa	Operation, method
सूत्र	sūtra	Rule, aphorism
भुजा	bhujā	Base (of triangle)
कोटि	koṭi	Perpendicular
कर्ण	karṇa	Hypotenuse, diagonal
श्रेणी	śreṇī	Progression, series
चय	caya	Common difference (in AP)
गुणोत्तर	guṇottara	Common ratio (in GP)
कुट्टक	kuṭṭaka	Pulverizer (linear Diophantine)
अङ्कपाश	aṅkapāśa	Combinatorial problems

iti līlāvatī samāptā

इति लीलावती समाप्ता ॥

Thus ends the Līlāvatī.